

Functional and Technical Requirements for Port of Dover System Dashboard

1. Functional Requirements

A. Real-Time Data Monitoring

Traffic Flow Visualization:

Display live traffic metrics (lorries, cars, coaches) across checkpoints (UK/French borders, ferry check-ins).

Track queue lengths, processing times, and buffer zone (BZ) status.

Vessel & Freight Tracking:

Monitor ship arrivals/departures, cargo types (refrigerated, general), and ferry schedules.

Highlight delays or bottlenecks in cargo handling.

Environmental Conditions:

Integrate weather data (e.g., wind speed, visibility) impacting port operations.

Resource Availability:

Show real-time staffing levels (stewards, supervisors), lane availability, and equipment status (e.g., PAF booths).

B. Operational Control

Lane Management:

Open/close lanes dynamically based on congestion (e.g., Flexi lanes for freight/tourist switching).

Automated alerts for lane overloads or checkpoint delays.

Steward & Resource Deployment:

Assign stewards to incidents or high-traffic zones via drag-and-drop interfaces.

Track resource utilization (e.g., "Deploy Stewards" button with GPS tracking).

Incident Response:

Log incidents (accidents, equipment failures) with severity levels.

Assign tasks to teams and track resolution progress.

Communication Hub:

Integrated chat/alert system for coordination with ferry operators, border agencies, and emergency services.

C. Security Management

Surveillance Integration:

Live feeds from CCTV cameras at critical zones (e.g., buffer zones, boarding ramps).

AI-driven anomaly detection (e.g., unattended cargo, unauthorized access).

Access Control:

Role-based permissions (e.g., operators vs. supervisors).

Two-factor authentication for sensitive actions (e.g., lane closures).

Data Protection:

Encrypt passenger/cargo data to comply with GDPR.

Audit trails for user actions (e.g., who modified lane settings).

D. Analytics & Reporting

Real-Time Dashboards:

Heatmaps for traffic congestion predictions (e.g., 24-hour forecasts).

Key Performance Indicators (KPIs): Queue times, vessel turnaround rates, incident resolution times.

Historical Analysis:

Compare current operations with historical trends (e.g., peak vs. off-peak performance).

Exportable reports for audits or strategic planning.

Predictive Modeling:

AI-driven insights for resource allocation (e.g., optimal steward deployment during cruise ship arrivals).

3. Example Use Cases

1. Priority Unloading of Perishable Goods

Scenario:

A refrigerated cargo ship carrying 335,000 tonnes of perishable goods (e.g., seafood, pharmaceuticals) arrives, but delays risk spoilage.

User Role: Cargo Supervisor

Dashboard Actions:

Priority Tagging: System flags the shipment as "High Priority" based on cargo type and shelf life.

Dynamic Queue Adjustment: Overrides standard queues to fast-track unloading.

Resource Allocation: Assigns additional cranes and cold-storage forklifts to the ship's dock.

IoT Integration: Monitors real-time container temperatures; alerts if thresholds are breached.

Outcome:

Perishable cargo unloaded in 2 hours (vs. standard 4 hours), preventing £1.2M in losses.

2. Oversized Cargo Handling

Scenario:

A ship arrives with wind turbine blades (oversized cargo) requiring specialized equipment and lane clearance.

User Role: Terminal Control Officer

Dashboard Actions:

Route Planning: Maps a dedicated corridor from the dock to storage, avoiding narrow lanes.

Equipment Scheduling: Reserves heavy-lift cranes and wide-load transporters.

Traffic Mitigation: Temporarily reroutes standard freight vehicles to alternate lanes.

Outcome:

Oversized cargo unloaded without disrupting general operations.

3. Queue Mitigation for High-Priority Ferry Departures

Scenario:

A ferry bound for urgent EU medical supplies must depart in 1 hour but is delayed due to cargo backlog.

User Role: Operations Manager

Dashboard Actions:

Priority Override: Assigns the ferry to a "VIP Lane" with dedicated loading crews.

Real-Time Queue Dashboard: Identifies bottlenecks in cargo screening and reroutes non-critical shipments.

Automated Alerts: Notifies border agencies to expedite customs clearance.

Outcome:

Ferry departs on time with 100% cargo loaded.

4. Dynamic Cargo Prioritization During Strikes

Scenario:

A labor strike reduces workforce by 50%, causing delays in cargo handling.

User Role: Terminal Control Manager

Dashboard Actions:

AI-Driven Priority Sorting: Ranks cargo based on:

Ship departure schedules.

Cargo type (e.g., medical supplies > general freight).

Contractual penalties for delays.

Resource Reallocation: Redirects remaining staff to high-priority docks.

Stakeholder Communication: Shares revised ETAs with logistics partners via the dashboard.

Outcome:

80% of high-priority cargo processed despite reduced staffing.

5. Hazardous Material Handling

Scenario:

A shipment of flammable chemicals requires strict safety protocols, slowing down unloading.

User Role: Safety Officer

Dashboard Actions:

Safety Zone Activation: Isolates the dock area; pauses non-essential traffic nearby.

Protocol Checklists: Guides staff through step-by-step hazardous material handling via the dashboard.

Real-Time Air Quality Sensors: Monitors for leaks and triggers alarms if unsafe levels detected.

Outcome:

Hazardous cargo unloaded safely with zero incidents.

6. Buffer Zone Optimization for Peak Traffic

Scenario:

A surge in lorry arrivals overwhelms the buffer zone, risking spillover onto local roads.

User Role: Traffic Controller

Dashboard Actions:

Predictive Buffer Management: Uses historical data to forecast peak times and pre-allocate buffer lanes.

Automated Diversion: Reroutes non-urgent freight to overflow parking zones.

Dynamic Signage: Updates digital signs to direct drivers to available slots.

Outcome:

Buffer zone capacity maintained; no traffic spillover.

7. Just-in-Time Loading for Time-Sensitive Shipments

Scenario:

A "just-in-time" automotive parts shipment must load onto a departing vessel within 30 minutes.

User Role: Logistics Coordinator

Dashboard Actions:

Live Shipment Tracking: GPS pins the truck's location and ETA to the port.

Dock Reservation: Reserves a loading bay closest to the vessel's cargo hold.

Priority Green Lane: Clears customs checks in advance using pre-submitted digital manifests.

Outcome:

Parts loaded with 5 minutes to spare, avoiding factory production halts.

8. Automated Crane Scheduling

Scenario:

Multiple ships arrive simultaneously, creating competition for cranes.

User Role: Crane Operator Supervisor

Dashboard Actions:

AI-Powered Crane Allocation: Assigns cranes based on:

Ship size/type.

Cargo urgency.

Crane proximity.

Conflict Resolution: Flags overlapping schedules and proposes alternatives.

Outcome:

90% crane utilization rate with zero downtime.

9. Post-Brexit Customs Bottleneck Mitigation

Scenario:

New customs checks post-Brexit slow down EU-bound freight processing.

User Role: Customs Officer

Dashboard Actions:

Digital Twin Simulation: Models checkpoint workflows to identify inefficiencies.

Pre-Clearance Portal: Allows drivers to submit documents in advance via the dashboard.

Dedicated EU Freight Lanes: Separates EU-bound cargo for faster processing.

Outcome:

Customs processing time reduced by 40%.

10. Emergency Rerouting During Natural Disasters

Scenario:

A storm damages a primary loading dock, forcing operations to shift to backup facilities.

User Role: Crisis Manager

Dashboard Actions:

Disaster Mode Activation: Suspends normal workflows and activates contingency plans.

Dynamic Dock Reassignment: Redirects incoming ships and lorries to undamaged docks.

Resource Tracking: Monitors backup generator status and fuel reserves in real time.

Outcome:

70% operational capacity maintained during recovery.

11. Peak Traffic Management:

Dashboard predicts a 2-hour delay at the French border using historical data and AI.

Terminal Control automatically reroutes freight to buffer zones and deploys extra stewards.

12. Incident Resolution:

A lane closure due to equipment failure triggers an alert.

Supervisors reallocate resources and notify ferry operators via the communication hub.

13. Executive Reporting:

Senior executives access a dashboard showing quarterly trade volume trends and environmental impact metrics.

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B. User Interface (UI/UX)

Role-Based Views:

Terminal Control Officers: Simplified interface with quick-access buttons (e.g., "Open Lane," "Alert Team").

Senior Executives: High-level KPIs (e.g., £144bn freight value, 33% UK-EU trade volume).

Accessibility:

High-contrast themes (dark/light mode) for control room lighting.

Voice command support for hands-free operation.

Customization:

Drag-and-drop widgets for personalized dashboards.

D. Future-Proofing

Modular Design:

Plug-and-play compatibility with future AI/ML models (e.g., predictive maintenance).

Support for 5G connectivity and edge computing.

API Ecosystem:

Open APIs for third-party developers (e.g., logistics companies).